A Hawkes Process Approach to NASDAQ Trade Flow:

AAPL execution data modelled via a bivariate, exponential-kernel Hawkes process: an evaluation of same-side vs cross-side excitation, and the relationship between execution volume & execution intensity.

**AAPL data accessed from:* lobster-data (n.d.) Hugging Face dataset repository. Available at: https://huggingface.co/datasets/totalorganfailure/lobster-data (Accessed: 04 August 2026).

Abstract

Equity market trade flow arrives in clusters rather than at a steady rate. Same and cross-side excitations in AAPL executions are estimated from a bivariate, exponential-kernel Hawkes process, validated via time-rescaling, and evaluated through branching ratios. The Spearman rank correlation coefficient between execution volumes and intensities is obtained for contemporaneous and lagged data pairings.

Both sell and buy streams rejected the Kolmogorov-Smirnov (KS) test with near-identical statistics (0.109 and 0.107 for sell and buy respectively), indicating the misspecification is not channel-specific; the ratio of same-side to cross-side branching ratio (14:1 and 16:1 respectively) is therefore unlikely to be affected by the misspecification. It should be noted, however, that the absolute branching ratios obtained are model-specific. The Spearman correlation coefficient showed negative and near zero values for contemporaneous data of sell and buy streams (-0.003 and -0.045 respectively), whereas lagged data demonstrated a weakly positive but statistically significant relationship (0.091 and 0.078 respectively) between volume and intensity.

From the large difference between same and cross-side influence, it can be observed that the order flow is largely direction preserved. Moreover, from the KS rejection, it can be concluded that the model failed to fit long interevent gaps, deviating greatly from the reference (y = x) beyond a theoretical quantile value of 3, meaning a long-range kernel would have been a better fit. The Spearman correlation coefficients pointed towards a correlation between previous execution sizes and subsequent execution intensities, whereas historic intensities cannot be said to correlate with subsequent volumes.

Theory & Code

The Hawkes process, introduced by Hawkes (1971), is a self-exciting point process whereby past event occurrences increase the intensity of future events and is represented as such:

$$\lambda(t) = \mu + \sum_{t_i<t} S(t) \quad (1)$$

In the AAPL message book, each execution's contribution to subsequent execution intensities decays exponentially. The excitation term in equation (1) is therefore presented as $\sum_{t_i<t} \alpha e^{-\beta(t-t_i)}$. The parameters that influence this process are estimated from the data via maximum likelihood estimation and comprise of four α values, representing same-side and cross-side execution jumps, one constant β, forcing all channels to decay at the same rate, and two μ values, representing the baseline event rate for sell and buy streams. These parameters are used to compare and evaluate same and cross-side excitation via branching ratios.

The utilisation of Ogata thinning enables a simpler and more exact Hawkes process. The algorithm draws τ (candidate arrival times) from an exponential distribution with rate λ^* (upper-bound intensity) with each candidate's acceptance probability being $\lambda(t)/\lambda^*$ (current candidate intensity over upper-bound intensity). Each candidate is then assigned an independent uniform draw and is accepted if the value drawn is below the probability threshold. For the chosen model, the initialised λ^* is the sum of the baseline rate of events for both sell and buy streams, and subsequent λ^* are a pool of λ^* from both streams. The acceptance probability of each stream is thus their respective $\lambda(t)$ over the pooled λ^*. For each stream, the cross-stream parameters are also updated e.g. a sell-buy which updates buy stream parameters within the sell stream conditional. The history of event times is then utilised to calculate a list of intensities for each stream that correspond to their respective event times via an intensity calculation function. This function evaluates each excitation kernel at every entry and therefore computes the intensity at every entry. Within the intensity calculation function, a failsafe is used to prevent the decay term becoming extremely large by converting all negative event times into infinite, causing the kernel to return exactly zero.

Since every term decays by the same factor over the same interval, a running-sum method is implemented for $S(t) = \sum_{t_i<t} \alpha e^{-\beta(t-t_i)}$ (accumulated excitation sum), resulting in a recursion:

$$S(t_n) = S(t_{n-1}) \times e^{-\beta(t_n-t_{n-1})} + \alpha \quad (2)$$

Equation (2) is used instead of recomputing the whole sum at each event, reducing the algorithm's cost to $O(n)$ from $O(n^2)$. Moreover, the compensator $\Lambda(t) = \int_0^t \lambda(s)ds$, the integral of conditional intensity, is evaluated in the form:

$$\Lambda(t) = \mu t + \sum_{t_i<t} \left(\frac{\alpha}{\beta}\right)(1 - e^{-\beta(t-t_i)}) \quad (3)$$

From equation (3), it can be observed that the term $\sum_{t_i<t} \left(-{}^{\alpha}/_{\beta}\right)(e^{-\beta(t-t_i)})$ can be written in the same recursion structure seen in equation (2) with increment ${}^{\alpha}/_{\beta}$. Therefore, equation (3) can be described as the sum of the baseline contribution (μt) and the sum of the integrated excitation kernel.

To recover, or rather estimate, these parameters from real data, a maximum likelihood estimate is used. The tutorial by Rizoiu et al. (2017) describes the log-likelihood, $logL(\theta)$, of a Hawkes process as such:

$$logL(\theta) = \sum_{i=1}^{N(T)} log\lambda(T_i) - \int_0^T \lambda(t)dt \quad (4)$$

Equation (4) is in the logarithmic form due to how small likelihoods become, avoiding the risk of running out of floating-point precision and preserving the location of the maximum. A log-space reparameterisation is conducted on the input parameters to return only positive parameters, preventing overflow of the exponential term and demonstrating a limiting assumption of the model: inhibition cannot be represented. To obtain the best parameters, the log-likelihood value is maximised via the 'minimize' function in the SciPy states module. This is accomplished by first minimising the log-likelihood values: negating them. The negated values are then fed into the function, causing the parameters with the largest log-likelihood values to be returned.

To validate the accuracy of the fitted model at the estimated parameters, time-rescaling is performed. Brown et al. (2002) show that point processes with integrable conditional intensities can be transformed into unit-rate Poisson processes. Therefore, the transformed processes have Exp(1) waiting times, so if the fitted model were accurate, the time-rescaled data points should lie on the line $y = x$ on a QQ plot. For the bivariate, exponential Hawkes process, the compensator is implemented to compute the rescaled interevent times for sell and buy streams, with the data then compared to Exp(1) using the 'kstest' function in the SciPy stats module.

Given the model neglects volume, treating volumes of drastically differing sizes as having the same influence, it is of interest to see whether there is a connection between execution volume and intensity. Execution volumes are heavily right-skewed. If Pearson's correlation coefficient were to be used, the small number of large execution volumes would dominate the relationship. Therefore, the Spearman rank correlation coefficient is the appropriate choice as it ranks each volume, weighting them equally.

Results

Fig. 1 – A plot of intensity as a function of time using synthetic parameters

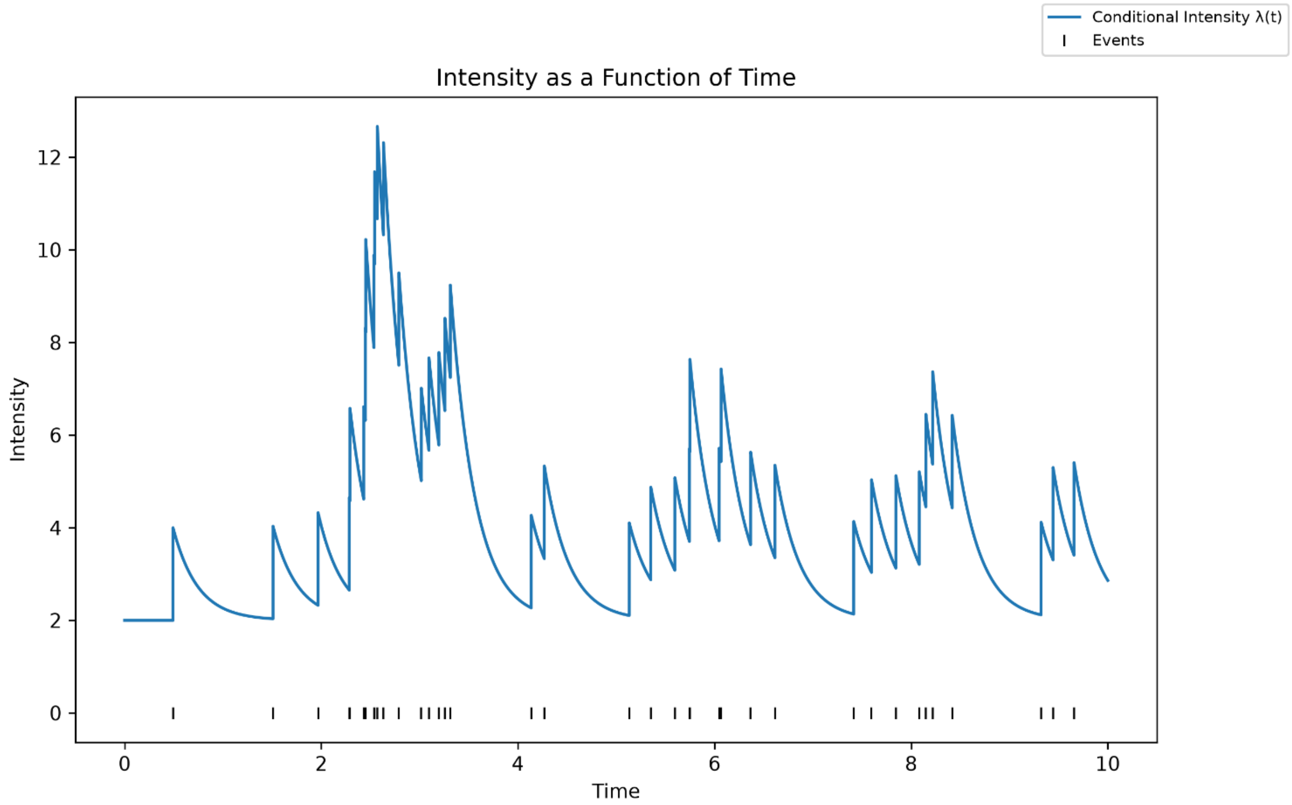


Fig. 2 – Percentage difference between recovered and synthetic parameters against max times

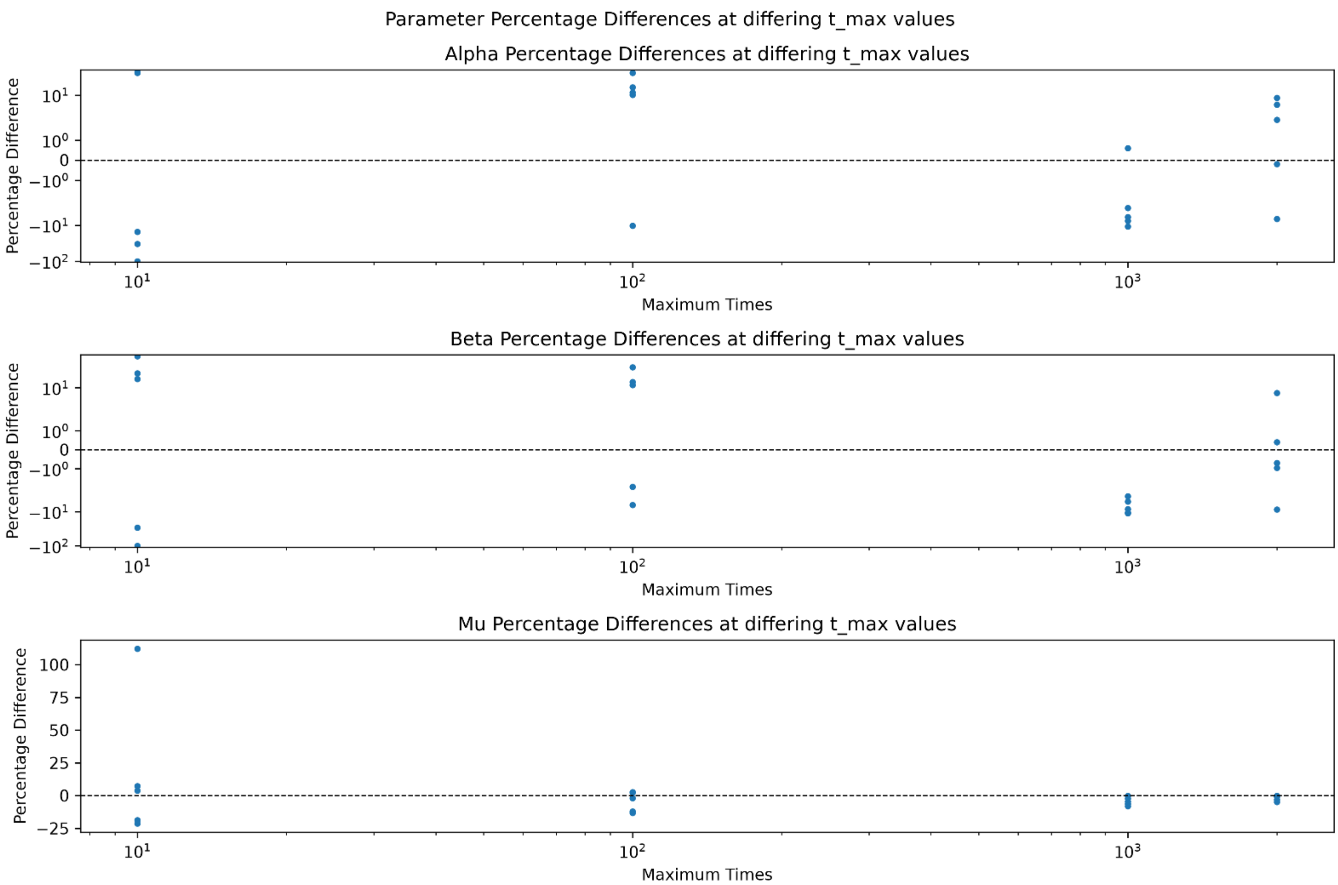


Fig. 3 – A plot of intensity as a function of time using AAPL Message Book data

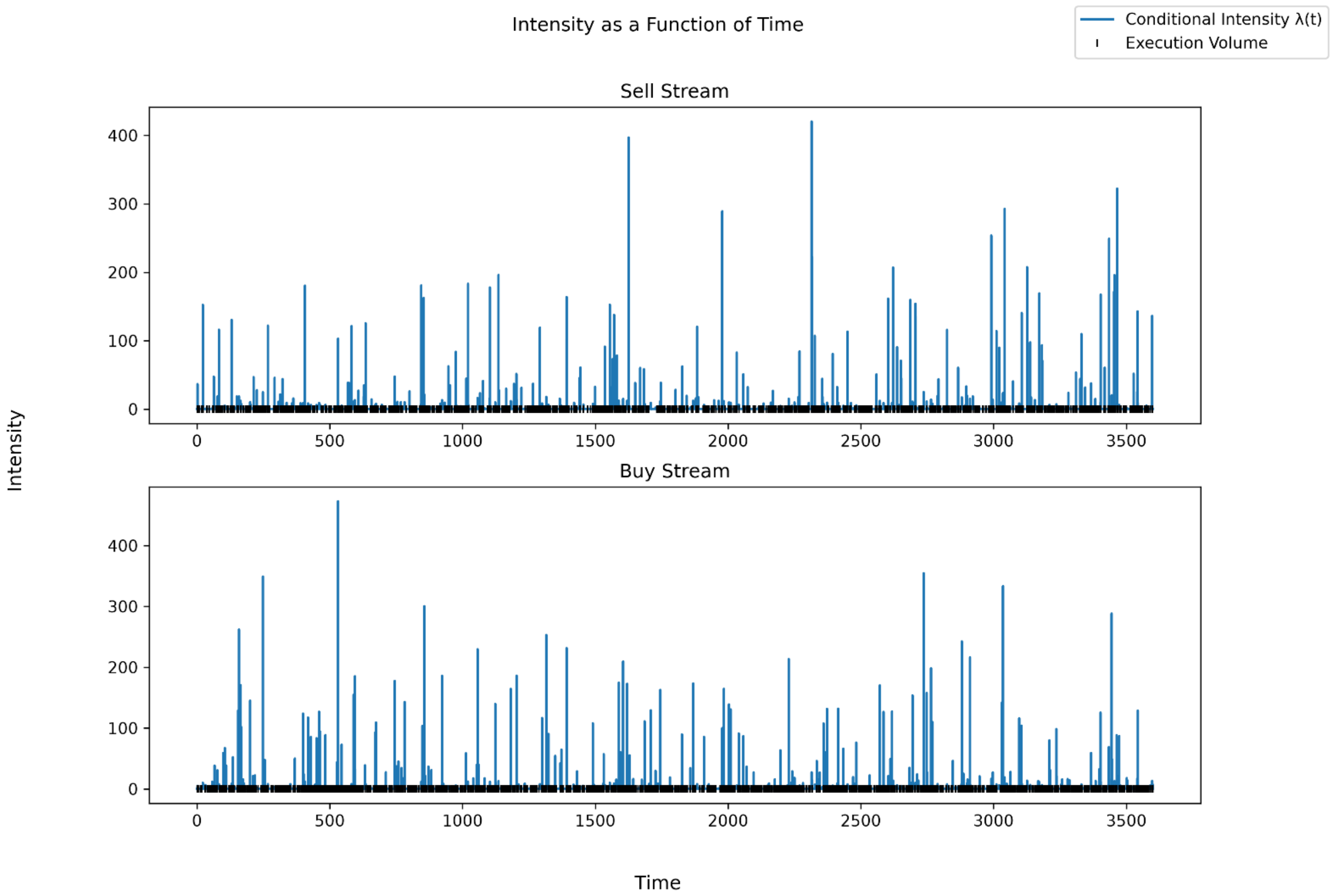


Fig. 4 – A QQ plot of time-rescaled inter-event intervals against the Exp(1) distribution for sell and buy stream

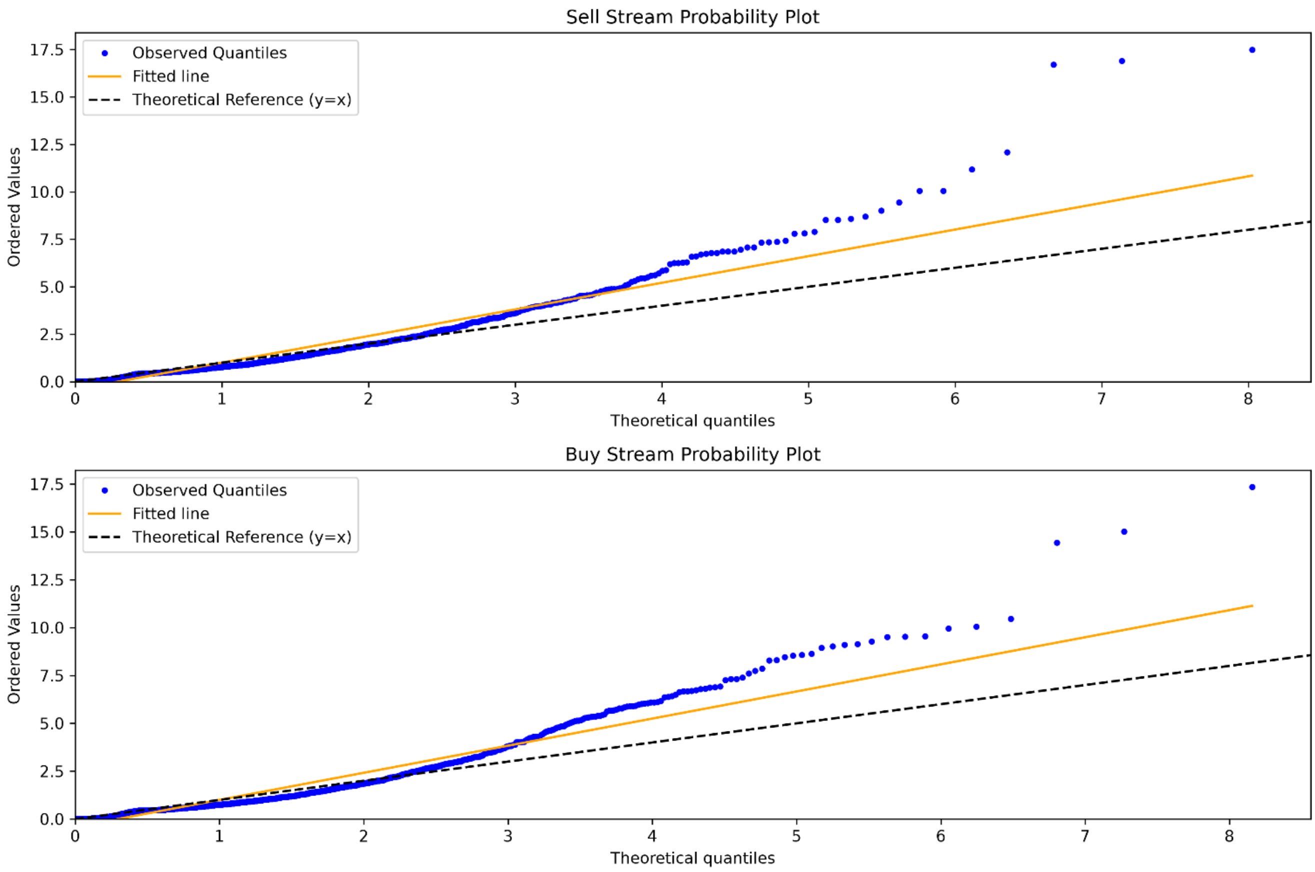


Fig. 5 – A chart of branching ratios of each execution type

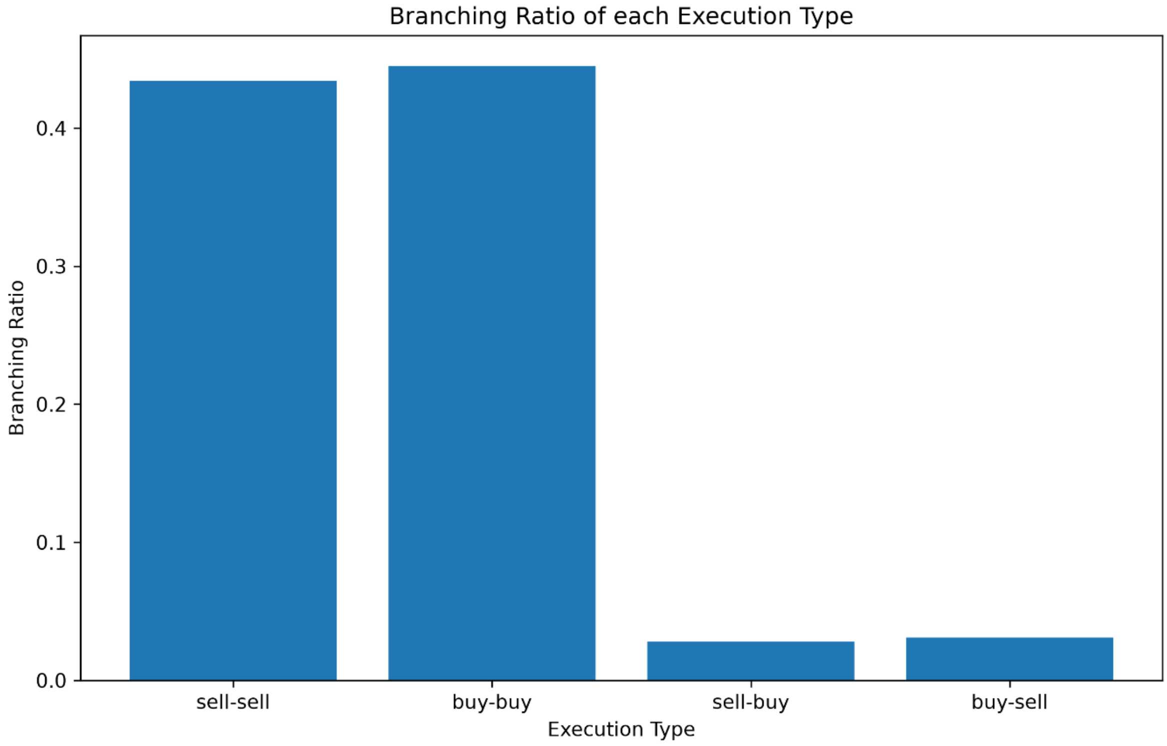


Fig. 6 – A plot of contemporaneous execution volume and intensity

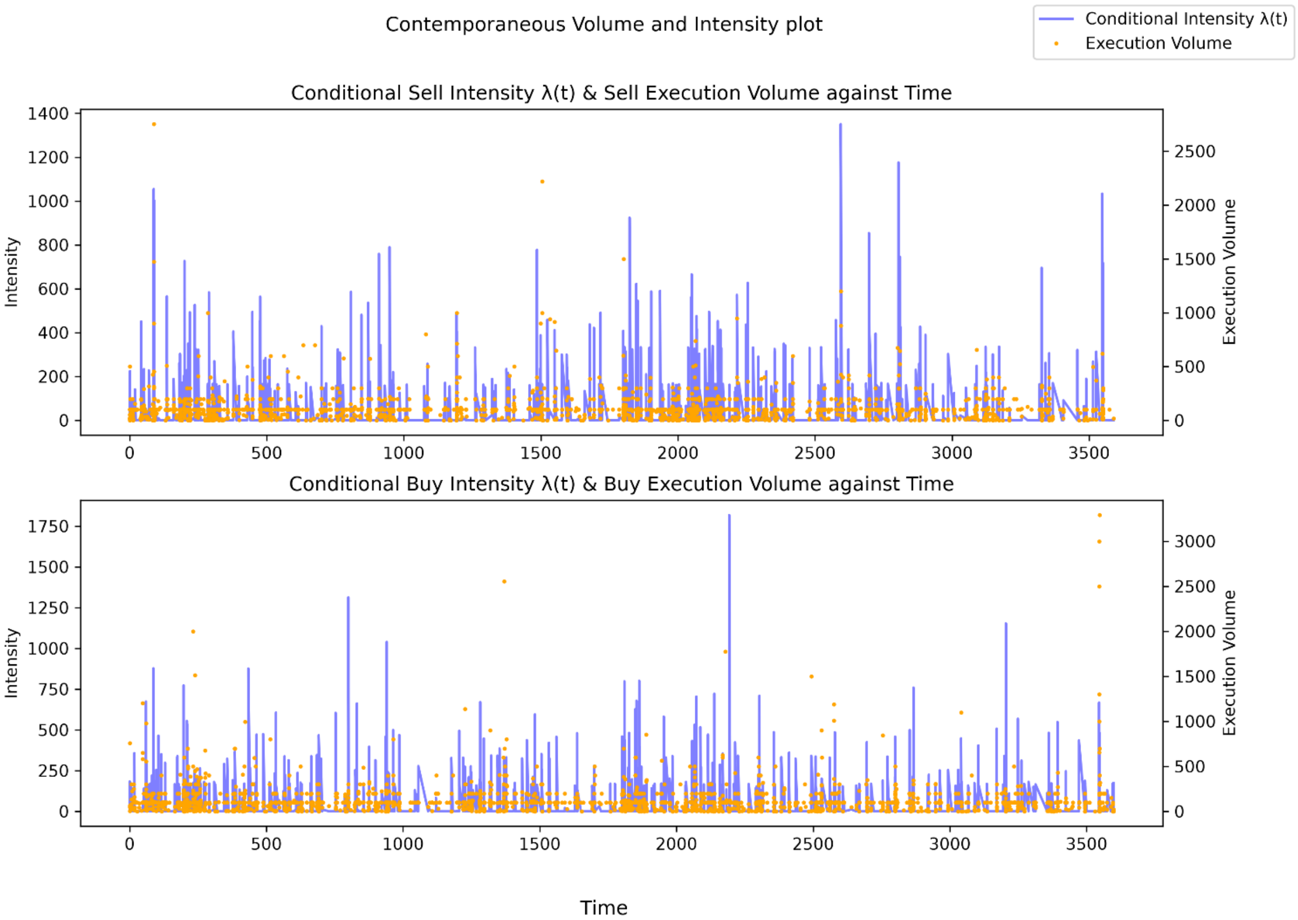
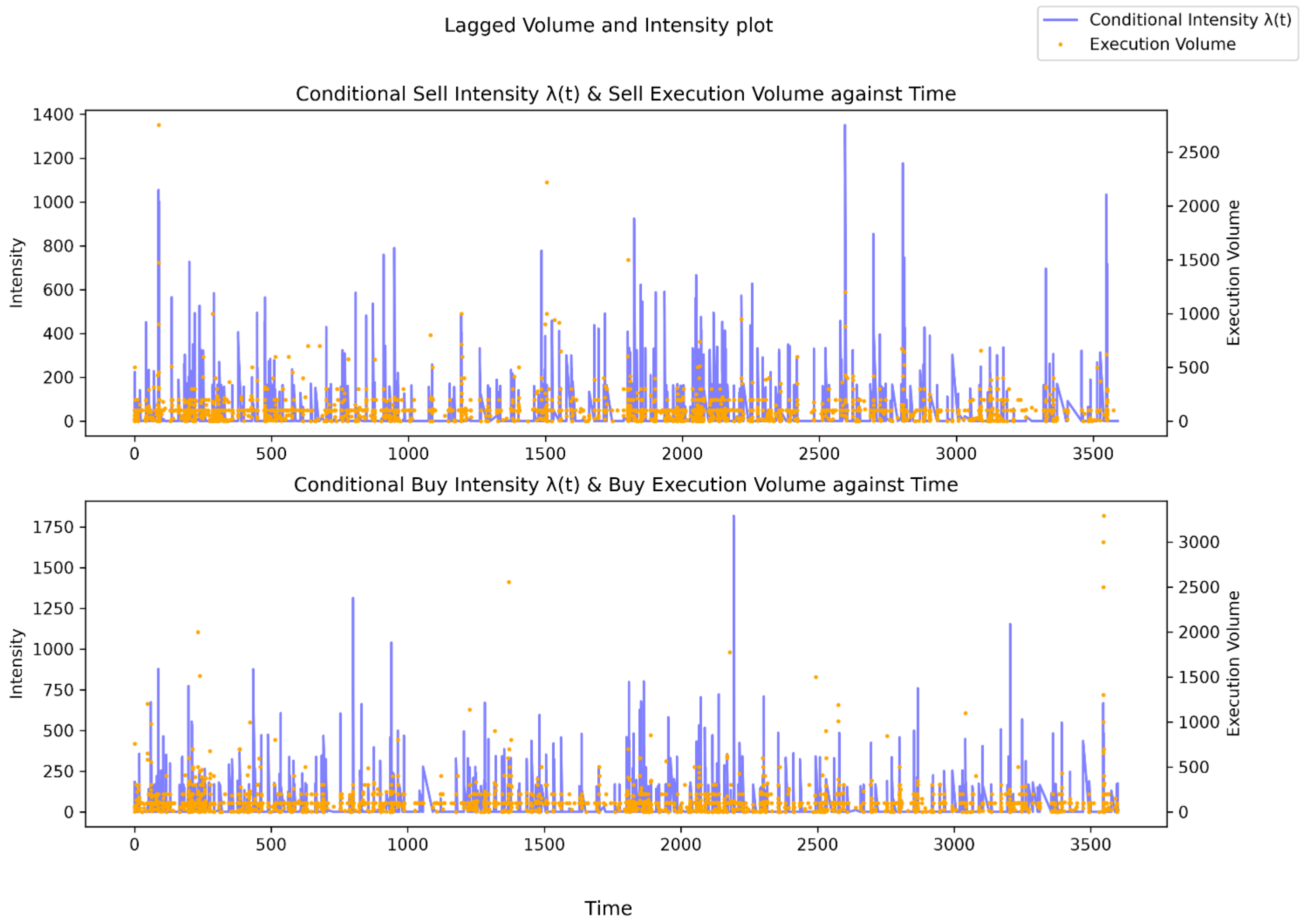


Fig. 7 – A plot of lagged execution volume and current execution intensity



KS Test results

	KS Statistic	P-Value	Statistic Location	Statistic Sign
Sell Stream	0.109	$2.418e^{-22}$	0.084	+1
Buy Stream	0.107	$1.312e^{-24}$	0.810	+1

Contemporaneous execution volume and intensity Spearman coefficients

	Rank Correlation Coefficient	P-Value
Sell Stream	-0.003	0.899
Buy Stream	-0.045	0.026

Lagged execution volume and intensity Spearman coefficients

	Rank Correlation Coefficient	P-Value
Sell Stream	0.091	$2.842e^{-5}$
Buy Stream	0.078	$1.145e^{-4}$

**Note that results for Fig. 1 and Fig. 2 are subject to change due to the random nature of the process.*

**Note that for Fig. 6 and Fig. 7 the one-event lag is not visually resolvable at this timescale. The difference appears in the Spearman Rank correlation coefficients reported in the Analysis.*

Analysis

It can be observed from Fig. 2 that all recovered parameters converge towards the original parameters as maximum runtime increases, with μ showing the greatest convergence. This is unsurprising given that the baseline event rate is embedded into the process as shown in equation (1). Therefore, μ contributes to all events. However, for α and β, they belong in the excitation term, meaning to extra the timing structure of the process must be analysed: whether events cluster, how tightly they cluster, and the rate at which the clusters decay. This difference between μ and α and β is clearly demonstrated in the percentage difference at t_max = 10, where the spread of α and β are significantly greater than that of μ precisely due to the limited data set that can be analysed to produce a suitable α and β.

From the KS Test results, for each stream, at the x position 0.084 for sell and 0.810 for buy, the difference between the observed proportion (from data used) and the theoretical proportion is greatest and is about 11% for both streams. The null hypothesis is that rescaled interevent times are drawn from Exp(1). The p-values demonstrate that the data, given the null hypothesis is true, is near 0, resulting in a decisive rejection. Since this is the case for both sell and buy streams and near identical, it can be said that the rejection does not stem from directional issues. Fig. 4 demonstrates the deviation of the rescaled interevent times from the theoretical line y=x, deviating from quantile 3 onwards. The deviating tail suggests that the exponential kernel was not an appropriate fit for, with the deviation direction indicating an overestimate after quantile 3. According to Hardiman, Bercot, and Bouchaud (2013), a better fit would be a power-law kernel due to the limitation of the exponential kernel: assumption of a single timescale β^{-1}.

Due to the rejection of the null hypothesis, the empirical branching ratio values can not be taken at face value. Therefore, the ratios between same and cross-side branching ratios for each respective stream is evaluated. Sell stream has a 14:1 ratio between same and cross-side branching ratio and 16:1 for the buy stream. This indicates that same-side executions tend to generate a greater average number of offspring events. It can be confidently concluded that same-side excitation is more dominant and can be observed the clearest.

Going from the contemporaneous relationship to the lagged, the Spearman coefficient of the relationship between execution volume and intensity shows, not only an increase, but also a clear change in sign. Moreover, the p-value, demonstrating the probability of no monotonic relationship, decreases significantly from contemporaneous to lagged further emphasises the stronger correlation in lagged data. Concretely, prevailing intensities are unable to predict future volumes, whereas execution volumes do relate to subsequent intensities.

Author's Notes

From this project, the specific model chosen was, although computationally better, not accurate for the data. As mentioned in the Analysis, a power-law kernel may have been better.

This project could have been extended to predict future intensities rather than be limited to the time duration, and expanding on strategic decision-making rather than solely focusing on an analysis of microstructures.

References

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